



2023 Andrei Borisovich Vistelius Research Award

Eric Grunsky¹ · Frits Agterberg¹

Published online: 17 August 2023

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Behnam Sadeghi

Dr. Behnam Sadeghi is a worthy recipient of the 2023 Andrei Borisovich Vistelius Research Award. As a scholar in mathematical geosciences, he has an impressive record of research and publications for an early career scientist. Behnam has developed a widespread network of research interests and affiliations with several universities and research institutes. Behnam has already made significant contributions to his field, with a number of high-quality publications including 40 papers in peer-reviewed journals and 30 conference proceedings, as well as one book, eight book chapters,

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and several research and industry reviews and reports. His presence on ResearchGate shows a steady increase in the site's RG Score and H-index.

In October 2011, while Behnam was pursuing his Master's degree, he had the opportunity to attend Frits Agterberg's workshop at a conference in Tehran, Iran. Behnam's thesis focused on the application and development of multifractal models in exploration geochemistry, and meeting Frits in person was a long-held aspiration of his. His M.Sc. thesis was awarded the university's best thesis of the year and was later published as a book in 2020. Ten years after their first meeting, Behnam and Frits collaborated on a chapter about Singularity Analysis for the Encyclopedia of Mathematical Geosciences. Behnam's talents and future potential were recognized in 2015 when he was granted the International Association for Mathematical Geosciences (IAMG) Student Travel Grant to attend the IAMG 2015 conference in Freiberg, Germany. Later, in 2022, he was also awarded the European Geosciences Union (EGU) General Assembly Roland Schlich Travel Support Grant.

Eric Grunsky served as the external examiner for Behnam's Ph.D. thesis, and he was impressed with Behnam's knowledge, deep understanding of quantitative theory, and skillful application of methods for mapping and predicting potential mineral prospects using multifractal and geostatistical methodologies in Sweden, Cyprus, and Turkey. Behnam's innovative application of fractal methods, along with geostatistical simulations and uncertainty quantification, has made a unique and significant contribution to the field of geosciences, with the potential for broader applications to other natural sciences and disciplines that rely on geospatial data. His Ph.D. thesis led to his recognition as the first IAMG awardee of the Founders Scholarship Award in 2020 that acknowledged his exceptional research efforts in the field of geomathematics. In recognition of his contributions, Behnam was invited to co-author several chapters for the Encyclopedia of Mathematical Geosciences, including one chapter about Uncertainty Quantification, with Eric Grunsky and Professor Vera Pawlowsky-Glahn.

Upon completing his Ph.D. studies at the University of New South Wales (UNSW), Australia, in 2020, under the supervision of Professor David Cohen and external supervisor Professor Jef Caers (Stanford University), he joined the EarthByte group at the University of Sydney (USYD) to work with Professor Dietmar Müller. Later, he transitioned to the mining and mineral exploration industry in Australia, where he served as a Senior Geoscientist. Despite his busy schedule, Behnam has remained committed to his research.

He has authored numerous book chapters and publications that include simulated-multifractal models and category-based and concentration–concentration (C–C) multifractal models. The C–C model was flagged by Frits as a very novel idea. Behnam also created novel decision-making algorithms for measuring uncertainty, and developed a Python package (TripleCpy) to apply the newly developed Chatterjee Correlation Coefficient in geosciences. Currently, Behnam is in the process of preparing a book on Fractals and Multifractals in the Geosciences, to be published by Elsevier and to be released in 2024.

Behnam's achievements in both academia and industry have earned him several prestigious awards and fellowships, including the Fulbright Research Fellowship (USA) in 2022. In addition, due to the novelty of his Ph.D. research, particularly in relation to GIS and mineral prospectivity mapping, uncertainty quantification, and

decision-making, he was selected as a top 4 and top 3 finalist in the Esri Young Scholars Award national competition in Australia in 2019 and 2020, respectively. He has also received Research Training Program (RTP) and postgraduate research scholarships (PRS) in Australia.

He is a Fellow of the Association of Applied Geochemists (FAAG), and the Australian Institute of Geoscientists (FAIG) (since 2022), and the Australian Association of Mining and Metallurgy FAusIMM—Fellow of the Australasian Institute of Mining and Metallurgy (since 2023).

Behnam's knowledge and productivity have earned him positions on the editorial boards of several highly ranked peer-reviewed journals, including *Ore Geology Reviews*, *Geochemistry*, *Minerals*, and *Earth Science Informatics*.

In 2023, Behnam commenced his position as an Impossible Without You (IWY) Research Fellow at CSIRO, Australia, in order to contribute to current and future projects with a particular focus on critical minerals. In parallel, Behnam will also complete his Fulbright Fellowship, splitting his time between the Carnegie Institution for Science's Earth and Planets Laboratory (EPL) with Dr. Shaunna Morrison, as the main host, and Stanford University with Professor Jef Caers.

Behnam is a highly enthusiastic, energetic, and self-motivated young researcher who has demonstrated exceptional leadership skills in various scientific conferences. For instance, he led a session at the Society for Geology Applied to Mineral Deposits (SGA) 2021 meeting (postponed to March 2022) in New Zealand, in which he worked alongside co-convenors Professor Franco Pirajno (Australia) and Professor Huayong Chen (China). He played a key role in organizing the Mineral Prospectivity and Exploration Targeting (MinProXT)-2022 conference and led a session at EGU 2023, "Data-driven discovery in mineralogy and geochemistry: data resources, analytics, and visualization." In that session, he collaborated with Drs. Shaunna Morrison and Aniruth Prabhu from the Carnegie Institution for Science, as well as Associate Professor Marshall Ma from the University of Idaho. Behnam is leading a session in IAMG 2023 with Professors Peter Atkinson (Lancaster University), Jennifer McKinley (Queen's University Belfast), and Mark Cooper (Geological Survey of Northern Ireland). The session is titled "Advances in computational analysis of geochemical survey data for geochemical anomaly detection." He is also providing a short course in IAMG 2023 with Putra Sadikin (IMDEX Pty Ltd) on Applied Geochemistry and Analytics. Behnam led a special issue of the journal *Geochemistry* in 2021, titled "Mineral exploration: a journey from fieldwork to lab work, computational modeling and mineral processing." He is currently leading a special issue for the same journal titled "Critical Minerals: Current Perspectives," 2023, with co-guest editors from Italy, India, and the USA. He is also leading one Special Issue for the journal *Minerals*, titled "Advanced Data Analytics in Earth and Environmental Sciences," with co-guest editors from Australia (USYD and Rio Tinto).

Behnam's research achievement, ongoing research, and his extensive research network clearly shows his multidisciplinary and interdisciplinary skill set in integrating GIS, data analytics, geostatistics, and mathematical geosciences in a highly applicable manner for mineral exploration. Behnam's academic achievements and productivity

are exceptional at this early stage in his career, and he is a worthy recipient of the Andrei Borisovich Vistelius Research Award.